MINJAE KWON

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Research Interests

Engineering nanostructures that shape light and fluidic behavior, guided by computational modeling.

Nanofabrication: periodic/3D nanostructures with tailored optical, mechanical, and interfacial properties.
Plasmonics & Metasurfaces: metasurface engineering for plasmonic field enhancement and optical sensing.
Electrokinetics & Simulation: multiphysics FEA optimizing electrokinetic particle capture and field enhancement.
Deep-Learning Readout: CNN-based image analysis for quantitative particle-capture recognition.

Education

Korea University (Seoul, Republic of Korea) Sep. 2025 – Present
Master's Student in the Department of Bioengineering
- Advised by Prof. Yong-Sang Ryu

Korea University (Seoul, Republic of Korea) Mar. 2023 – Aug. 2025
B.S. in Biomedical Engineering & Electrical Engineering (Double Major)
- GPA: N.A / 4.50

Konkuk University (Seoul, Republic of Korea) Mar. 2017 – Feb. 2023
Majored in Electronic and Electrical Engineering
- Transferred to Korea University
- Leave of absence due to national military service: Jul. 2019 – Jan. 2021

Research Experience

Ryu Group, Korea University (Seoul, Republic of Korea) Aug. 2024 – Present
Undergraduate/Graduate Student Researcher (Advised by Prof. Yong-Sang Ryu)
- **Current work: MIM Nanohole Plasmonic Metasurface for Single-Particle Analysis (in preparation)**
  Fabrication: Fabricating large-area nanohole structures via laser interference lithography (LIL), engineering a single-nanoparticle-capture geometry with a metal–insulator–metal dielectric stack to amplify SERS enhancement beyond prior work, enabling electrokinetic trapping of single particles at the hotspots.
  Simulation: Modeling optical (COMSOL ewfd) and electrokinetic (COMSOL ec/spf) fields to find geometries giving up to ~2×10^6 $|E/E_0|^4$ enhancement, and validating DEP/ACEO single-particle capture dynamics.
  CNN Analysis: Applying CNNs to SEM images to quantify the relative standard deviation (RSD) of single-nanoparticle capture to analyze biosensor uniformity and the resulting SERS signal.
- **Collaborative projects**
  PRET metasurface: Depositing capture structures onto the Plasmon Resonance Energy Transfer metasurface built by Prof. Junsuk Rho's group, enabling electrokinetic concentration of target analytes in aqueous solution.
  Hydrogen-sensing MEMS: Providing deposition and lithography support for a terahertz-resonant MEMS chip whose resonance shift enables label-free hydrogen sensing.
  Electrochemical-sensing nanostructure: Leading full-chip fabrication of a capture-free MIM nanohole variant, tuning hole size, nano-gap thickness, and materials for fast and sensitive electrochemical detection.
  KU–KIST joint project: Serving as a collaborating researcher at the Nanophotonics Research Center of KIST, developing a label-free plasmonic metasurface for sensing GPCR conformational dynamics.

Creative Challenger Program, Korea University Aug. 2023 – Feb. 2024
Project Leader, Vine Robot Colonoscopy Project (Advised by Prof. Daehee Hong and Dr. Sang-Hyun Kim)
- Developed and validated a novel soft-robot colonoscopy platform; conducted *in vivo* experiments in collaboration with the Department of Gastroenterology, Korea University Medical Center.
- Recognized with two Excellence Awards (15th Creative Challenger Program, 2nd Changemakers Forum)

Technical Proficiency

Fabrication & Characterization Skills

- Lithography: Photolithography, Laser Interference Lithography (LIL), Electron Beam Lithography (EBL), Soft Lithography, Nanoimprint Lithography (NIL)
- PVD (Physical Vapor Deposition): Thermal Evaporation, Electron Beam Evaporation, Sputtering
- CVD (Chemical Vapor Deposition): ALD, PECVD
- Etching: RIE, ICP-RIE
- Surface Treatment: Oxygen Plasma Ashing, UV-Ozone
- Material / Structure Analysis: SEM, AFM, Energy Dispersive X-ray Spectroscopy (EDS), Surface Profiler, Ellipsometry, Reflectance Spectroscopy, Raman Spectroscopy, Optical & Epi-fluorescence Microscopy

Computational & Design Skills

- Language: Python, C/C++, LaTeX, MATLAB
- Simulation: COMSOL Multiphysics (FEA), Lumerical FDTD
- Data Analysis: OriginLab, Veusz, Python-based data analysis (PyTorch, OpenCV)
- 2D/3D Design: Rhino, AutoCAD, KeyShot
- Physical Design: PCB/Breadboard Circuit Design, 3D Printing, Metal / Polymer CNC, Laser Cutting

Publications

[2] **M. Kwon**[‡], S. Lee[‡], S. Choi, D. Pyun, D.-H. Kim, M.-K. Kim[*], and Y.-S. Ryu[*], *"**Scalable nanohole plasmonic metasurface for electrokinetic single-particle capture and SERS analysis**,"* manuscript in preparation (2026).

[1] S. Lee[‡], S. Choi[‡], T. Kim, **M. Kwon**, T. J. Mun, S. Son, H. Seong, K. Lee, H. Im, E.-S. Yu, M.-K. Kim[*], and Y.-S. Ryu[*], ***"Analyte-accessible field volume as a governing design principle for nanophotonic sensing,"***
under revision, Advanced Materials (2026).

[‡]Equal contribution, [*]Corresponding author

Teaching Experience

Teaching Assistant, Guest Lecturer

Biomedical Semiconductor Process (BMED442, Prof. Yong-Sang Ryu)	Fall 2025, Fall 2026
Biomedical Design (HBL514, Prof. Yong-Sang Ryu)	Spring 2026

Awards and Scholarships (Selected)

BK21 FOUR Graduate Research Scholarship, Ministry of Education (Full Tuition)	Sep. 2025 – Present
Best Poster Presentation Award, Korean Biochip Fall Symposium	Nov. 2025
Outstanding New Student Scholarship, Korea University	Sep. 2025
Dean's List (Highest Honors, 2 semesters), Korea University	Mar. 2023 – Jun. 2025
Excellence Award, 2nd Changemakers Forum	Nov. 2024
Best Startup Award, Korea University	Nov. 2024
Third Place, Capstone Design Fair	Sep. 2024
Excellence Award, 15th Creative Challenger Program	Feb. 2024

Leadership & Extracurricular Activities (Selected)

School of Biomedical Engineering, Korea University

Founder & President, KU Biomedical Eng. Abroad Mentorship (KUBEAM) Aug. 2025 – Present

- Founded a mentorship initiative connecting 80+ students and alumni pursuing study/careers abroad, building a shared study-abroad resource hub and overseas alumni network.

Biosensor Team Lead Organizer, Biomedical Engineering Academic Team (BEAT) Aug. 2025 – Jul. 2026

- Led and mentored 60+ undergraduates in acquiring the concepts and technical skills necessary to explore the field of biosensors (e.g., SEM imaging, data analysis, critical literature review, presentation).

Senior Advisor & Vice President, Department Student Council Nov. 2024 – Nov. 2025

- Advised and supported planning and execution of academic, cultural, and community events for 500+ undergraduates. Led a 30+ member committee and secured $22,000 in sponsorships for a 500+ student alumni day.

65th Medical Brigade, 8th U.S. Army Jul. 2019 – Jan. 2021

Combat Medic & Medical Support Assistant, BDAACH (549th Hospital Center)

- Selected for the KATUSA (Korean Augmentation to the U.S. Army) program based on language proficiency. Served as staff in the Physical Performance Service Line (musculoskeletal rehabilitation and physical therapy) for outpatient care and on the COVID-19 Response Team (quarantine operations, vaccine administration).
- Served as Senior KATUSA of 65th Medical Brigade, the ranking KATUSA representative and liaison to U.S. command; received the Army Achievement Medal; honorably discharged as Sergeant (E-5).

Languages

Korean Native speaker
English TOEFL iBT 5.0 / 6.0 (Comparable 100 / 120)